Elementary row operations and elementary column operations on a matrix are known as elementary transformations.

We use the following notations for elementary row transformations:

(i) Interchanging of ith and j^{th} rows is denoted by $R_i \leftrightarrow R_j$.

(ii) The multiplication of each element of ith row by a non-zero constant λ is denoted by $R_i \to \lambda R_i$.

(iii) Addition to i^{th} row, a non-zero constant λ multiple of jth row is denoted by $R_i \to R_i + \lambda R_j$.

Similar notations are used for elementary column transformations.

Definition 1.4

Two matrices A and B of same order are said to be equivalent to one another if one can be obtained from the other by the applications of elementary transformations. Symbolically, we write $A \sim B$ to mean that the matrix A is equivalent to the matrix B.

For instance, let us consider a matrix $A = \begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix}$.

After performing the elementary row operation $R_2 \to R_2 + R_1$ on A, we get a matrix B in which the second row is the sum of the second row in A and the first row in A.

Thus, we get $A \sim B = \begin{bmatrix} 1 & -2 & 2 \\ 0 & -1 & 5 \\ 1 & -1 & -4 \end{bmatrix}$.

The above elementary row transformation is also represented as follows:

$$\begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix} \xrightarrow{R_2 \to R_2 + R_1} \begin{bmatrix} 1 & -2 & 2 \\ 0 & -1 & 5 \\ 1 & -1 & -4 \end{bmatrix}.$$

Note

An elementary transformation transforms a given matrix into another matrix which need not be equal to the given matrix.

1.3.2 Row-Echelon form

Using the row elementary operations, we can transform a given non-zero matrix to a simplified form called a **Row-echelon form**. In a row-echelon form, we may have rows all of whose entries are zero. Such rows are called **zero rows**. A non-zero row is one in which at least one of the entries is not zero. For instance, in the matrix $\begin{bmatrix} 6 & 0 & -1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$, R_1 and R_2 are non-zero rows and R_3 is a zero row.

Definition 1.5

A non-zero matrix E is said to be in a **row-echelon form** if:

(i) All zero rows of E occur below every non-zero row of E.

(ii) The first non-zero element in any row i of E occurs in the jth column of E, then all other entries in the jth column of E below the first non-zero element of row i are zeros.

(iii) The first non-zero entry in the ith row of E lies to the left of the first non-zero entry in $(i+1)^{th}$ row of E.

12th_Maths_EM_Vol1_CH 1_Matrices.indd 17 16-12-2022 12:16:59